

Course COMP-2560
Instructor Dr. B. Boufama
Assignment 04
Due date March 7, 11.59pm

Using system calls, fork(), wait(), open(), close(), read() and write(), write a C program to process a sentence by changing any uppercase letter into lowercase and vice versa. In particular, your program should

- Prompts the message "This program is a text processing tool",
- Gets in an infinite loop then
 1. Prompts the message "Enter a sentence, e.g., Good day, my name is BigFoot",
 2. Reads a whole line from the keyboard,
 3. Uses system call fork() to fork then,
 - the parent writes the message "I just created a child to perform a specific task for me, waiting for my child... then, calls wait() to wait for its child process.
 - the child process calls function childFunction()
 4. The child, through childFunction(char *line),
 - writes the message "I am a child process working for my parent"
 - codes the text stored in line (E.g, the above sentence becomes "gOOD DAY, MY NAME IS bIGfOOT"),
 - in case of an empty line, the child calls exit(55),
 - otherwise, it opens the file called code.txt, for writing (truncates it if it exists to get rid of old contents),
 - writes the coded sentence in code.txt, then closes the file.
 - calls exit(0)
 5. Once the child terminates, the parent checks the returned status value and if it is 55, writes on the screen "Error, sentence missing" and goes back to 1.
 6. Otherwise, the parent opens file code.txt for reading, reads the coded sentence, prints it on the screen and, closes code.txt before going back to 1.

important:

1. All reads/writes must be done using system calls read()/write()
2. Any system error should be handled and an appropriate error message should be displayed using **perror()**.
3. Assignments should be uploaded to Brightspace as a text file (.txt).
4. You can get some help from the TAs, consult their schedule on Brightspace.